

Practice Quiz 01A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Truth tables and logical equivalence.

1. Use a complete four-row truth table to determine whether

$$P \wedge (P \vee Q) \quad \text{and} \quad P \wedge Q$$

are logically equivalent. State your conclusion and identify a row supporting it if the formulas are not equivalent.

Solution

P	Q	$P \vee Q$	$P \wedge (P \vee Q)$	$P \wedge Q$
F	F	F	F	F
F	T	T	F	F
T	F	T	T	F
T	T	T	T	T

The formulas are not logically equivalent. When $P = T$ and $Q = F$, the first formula is true and the second is false. Logical equivalence requires agreement in every row. ■